

TABLE OF CONTENTS

TABLE OF CONTENTS	1
1- OVERVIEW	2
1-1 Tools and Instruments Required.....	3
2- INITIAL CONDITIONS	4
3- HARDWARE PREPARATION	5
4- HEAD CHECKS	9
5- BODY CHECKS	19
6- AMPLIFIER SHUTDOWN VERIFICATION	29
7- DECLARATION FORM PREPARATION	33
8- SYSTEM RESTORATION	33
APPENDIX A —SIGNAL POWER MONITOR FUNCTIONAL TEST DECLARATION FORM	34
APPENDIX B — RF POWER MEASUREMENT KIT USE	35
B-1 RF Formulas and 1.5T Division Conversions.....	35
B-2 RF Power Measurement Kit Use with 1.0T Systems — Recalculation.....	36
APPENDIX C — POWER MONITOR MEASUREMENTS	37
C-1 Tutorial example for 1.5T and 1.0T Head Test (Peak Power <i>High</i>):	37
C-1-1 Optional: VPP Ranges for 1.5T and 1.0T Head Test (Peak Power <i>High or Low</i>):	38
C-2 Tutorial example for 1.5T and 1.0T Head Test (Pulse Width PW <i>High, Min Limit</i>):	39
APPENDIX D — CALIBRATE SCOPE USING +4dBm SCOPE CALIBRATOR — 1.5T ONLY	41
REVISION HISTORY	42

1- OVERVIEW

This section (written as a continuous procedure) applies to systems installed with the RF/PEN 1 Cabinet (contains the RFSC and the MEPS).

This procedure ensures that the power monitor is fully functional, including redundant trip capability. The tests described use a special pulse sequence data base (PSD) to provide excitation for the RF amplifier. The PSD allows the service engineer to selectively force peak power, pulse width, and duty cycle beyond allowable limits. Setpoints for testing are selected by modifying control variables (CVs) that override the values that are normally calculated and downloaded by the PSD.

Power Monitor firmware operation is changed to bypass/test mode by moving jumpers on the Communications Manager/Power Monitor Board (JP2, JP6). This action permits setpoint verification without tripping the RF amplifier, thereby eliminating the time consuming necessity of resetting the amplifier after each trip.

After setpoints are checked, actual shutdown of the amplifier is verified twice for monitor A (to verify both relay and logic shutdowns), and once for monitor B (connected to amplifier via monitor A). This also confirms that the bypass/test jumper (JP2, JP6) is restored prior to returning the system to normal operation.

1-1 Tools and Instruments Required

Oscilloscope: 100 MHz minimum.

Additional tools per Table 1-1 or Table 1-2.

TABLE 1-1
TOOLS AND INSTRUMENTS REQUIRED (1.0 T AND 1.5T) LX

Item	Description	Part Number	Quantity
A.	RF Power Measurement Kit	46-317724G1 or G2	1
OR	G1 Kit does not contain the Dummy Load: 50 ohm, 200 Watt, 30dB Attenuator - Bird Model 8322	46-255837P10	

TABLE 1-2
TOOLS AND INSTRUMENTS REQUIRED (1.0 T AND 1.5T) LX

Item	Description	Part Number	Quantity
A.	RF Wattmeter - Bird Thruline Model 4391	46-255837P1	1
B.	10000-watt interchangeable element Bird # 10000B, 1.5T	46-255837P101	1
C.	5000-watt interchangeable element Bird # 5000B, 1.5T	46-255837P105	1
D.	RF Test Cables Kit	46-255816G1	1
E.	50 ohm, 200 Watt, 30dB Attenuator - Bird Model 8322	46-255837P10	1
F.	Attenuator, Fixed 50 ohm, 10 dB	46-258601P3	1
G.	Variable Rotary Attenuator, 10 dB, in 1 dB steps	46-255838P3	1 or 2

2- INITIAL CONDITIONS

Prerequisite calibration procedures listed in Table 2-1 must have been performed in sequence.

TABLE 2-1
PRE-REQUISITE — **POWER MONITOR CALIBRATION PROCEDURE LX**

Step	Procedure	When Required
1a OR	DUMMY LOAD & CABLES CALIBRATION	When not using the RF Power Measurement Kit and dummy load and cables have not been calibrated.
1b	RF POWER MEASUREMENT CALIBRATION	1.5T Only: Oscilloscope display (less than 300 MHz) must be characterized by performing calibration procedure using the 1.5T Scope Calibrator to compensate for bandwidth limitations (scope roll-off).
2	APB CALIBRATION MAX POWER RF OUT SETUP & CALIBRATION	Initial installation or if RF Amp has been repaired.

Pre-requisite — Signa must be fully functional and properly calibrated.

Pre-requisite — Software must be running with the system capable of normal imaging.

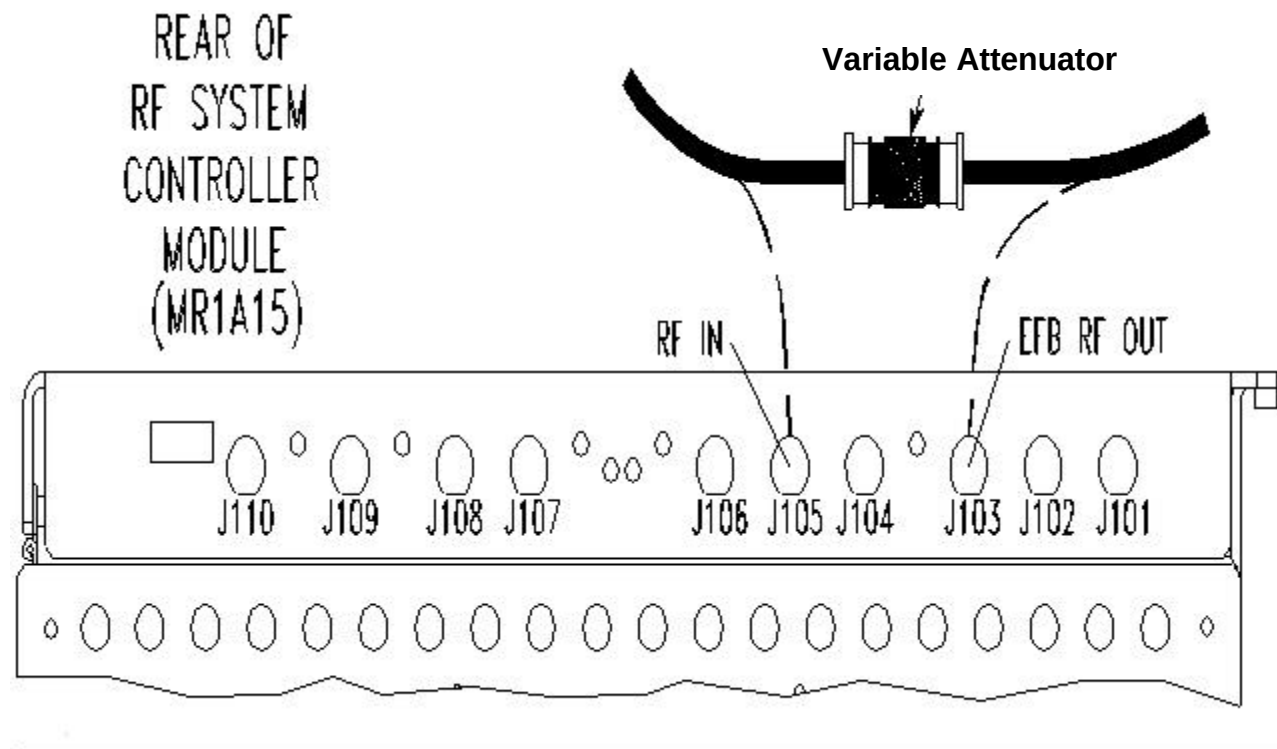
3- HARDWARE PREPARATION

1. Bypass EFB with a variable attenuator between RFIN, MR1A15J105, and EFB RF OUT, MR1A15J103, on the rear panel of the RF System Controller Module (RFSC). See Illustration 3-1.

Note

EFB is bypassed (with added attenuation) due to the use of square waveforms in this test (spikes on the front end of the RF square pulse could result in false trips).

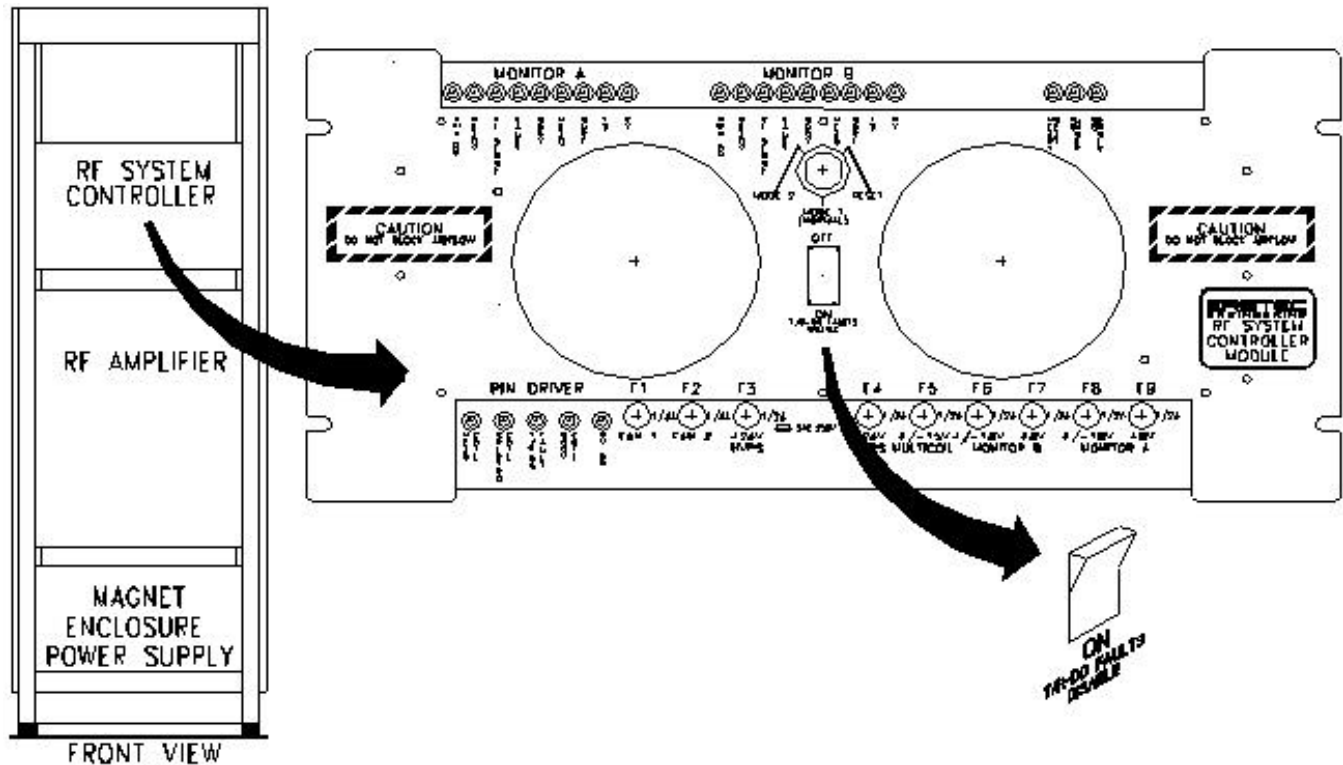
1. Variable Attenuator set-point:
 - A. Signa Advantage and Horizon Systems — set to 6 dB.
 - B. Lx / 8x Systems — set to 2 dB.



BYPASSING EFB
ILLUSTRATION 3-1

2. Remove the front cover from the RF/PEN cabinet.
3. Be sure that the monitor interlock switch on the front panel of the RFSC is in "MODE 1" (normal) position.

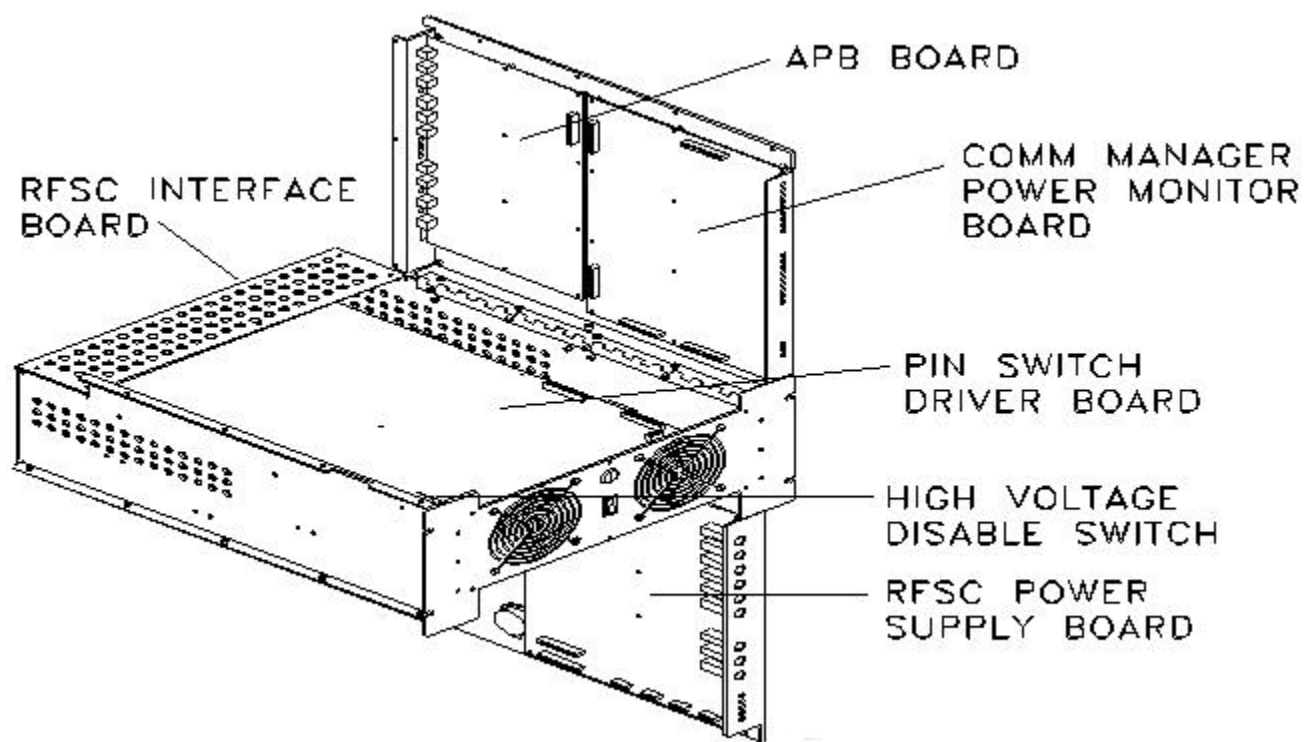
4. Place the rocker switch on the front panel of the RFSC Module in the ON (T/R-DD Faults Disable) position (bypass/test). The switch color will be red. See Illustration 3-2.



TR AND DD FAULTS DISABLE SWITCH POSITION AT RFSC MODULE
ILLUSTRATION 3-2

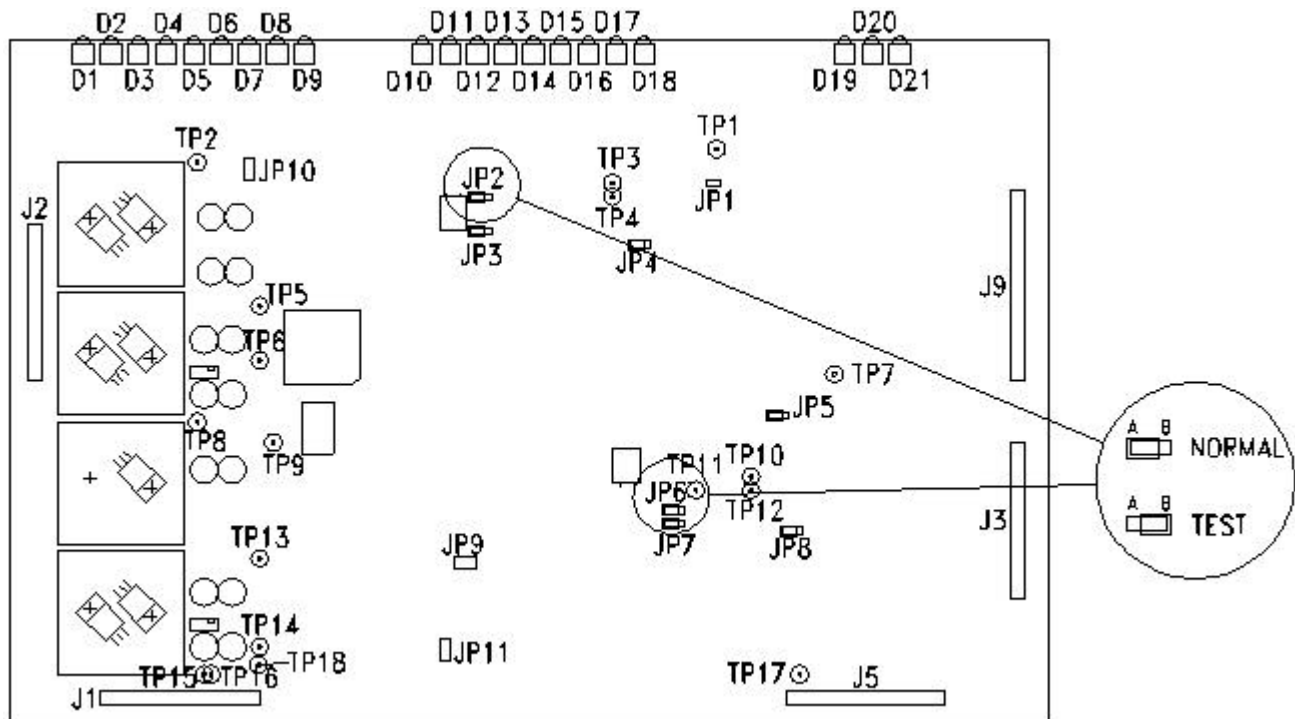
5. Remove the four 10-32x1/2 panhead screws and four #10 flat washers securing the RFSC (MRIA15) to the RF/PEN cabinet, and slide out the module.

6. Loosen the captive screws on the top lid and open. See Illustration 3-3. Do not open bottom cover.
7. Most RFSC Modules have a white High Voltage Enable Switch that must be pulled to the UP position (located near left front of module).



RF SYSTEM CONTROLLER MODULE
ILLUSTRATION 3-3

8. Place jumpers JP2 & JP6 on the Communications Manager/Power Monitor (CM/PM) board in the bypass/test position, "B". See Illustration 3-4.



CM/PM BOARD POWER MONITOR JUMPER SETTINGS

ILLUSTRATION 3-4

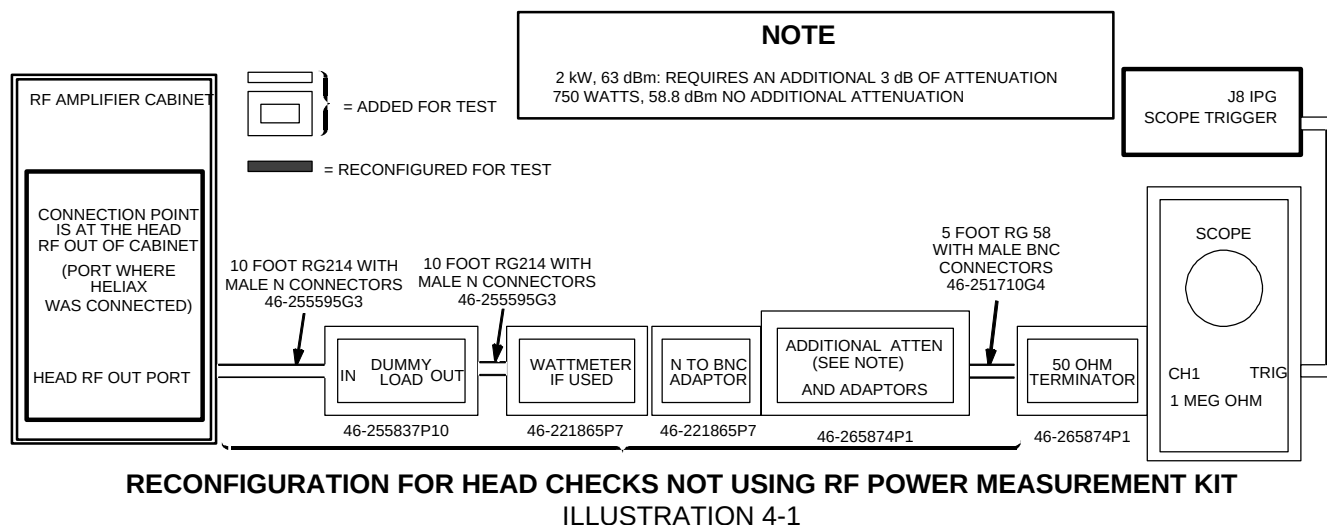
Note

Amplifier Tripped While Moving JP2 And JP6: If the amplifier is tripped while moving JP2 and JP6, IPER (In Place Error Recovery) should reset it about three minutes after the fault condition is cleared. If problems are encountered with amplifier operation after such a trip, rebooting Signa may be required to restore proper operation.

9. Verify that only the "Power On" LEDs are illuminated for both power monitors (Ready LED off, confirm that JP2 & JP6 are in test/bypass "B" position).

4- HEAD CHECKS

The following generic Illustration 4-1 has been provided for sites without the RF Power Measurement Kit.



Note

Maximum RF Signal input for any scope is 30 dBm (10 Volts Peak, 20 VPP).

Note

Both the dummy load and the cable that connects the amplifier output to the dummy load input must be calibrated per *DUMMY LOAD AND CABLES CALIBRATION*. See Signa Release 8.X Service Methods CD-ROM. If the cable has not been calibrated, STOP; do not continue with this procedure until cable calibration is complete.

1. **For 1.5T systems:** Connect per Illustration 4-1 to Head output (J12 at I/F panel / J2 at Head Directional Coupler) of RF/Pen cabinet. If using a wattmeter use the proper element. Or use the RF Power Measurement Kit card # 63 (2 kW, 63 dBm). For 1.5T is necessary to use the 1.5T Scope Calibrator card to compensate for scope bandwidth limitations (frequency roll-off), refer to the Appendix for instructions.

Verify the selectable oscilloscope input termination is 1 Meg ohm when using the RF Power Measurement Kit 50 ohm terminator. Otherwise, the oscilloscope should be 50 ohm termination for all RF measurements.

Note

Calibrate The Amplifier Output To Wattmeter Input Cable - The cable that is used to connect the amplifier output to the wattmeter input must be calibrated per procedure Dummy Load and Cables Calibration. If the cable has not been calibrated, STOP, do not continue with this procedure until ATTEN TEST cable calibration is complete.

Note

A tool is available for use in converting between the following: dBm, VPP, Watts, and Divisions (Divisions is used with 1.5T systems only). This tool is located on the Signa Release 8.X Service Methods CD-ROM under the Tools directory, titled: SCOPECAL.xls.

Path:

[START] [Programs] [Windows Explorer] [Mrservice (drive)] [Tools] [SCOPECAL.xls].

2. **For 1.0T systems:** Connect per Illustration 4-1 to Head output (J12 at I/F panel / J2 at Head Directional Coupler) of RF/Pen cabinet. If using a wattmeter use the proper element. Or use the RF Power Measurement Kit card # 58 (750 W, 58 dBm).

Verify the selectable oscilloscope input termination is 1 Meg ohm when using the RF Power Measurement Kit 50 ohm terminator. Otherwise, the oscilloscope should be 50 ohm termination for all RF measurements.

Note

Calibrate The Amplifier Output To Wattmeter Input Cable - The cable that is used to connect the amplifier output to the wattmeter input must be calibrated per procedure Dummy Load and Cables Calibration. If the cable has not been calibrated, STOP, do not continue with this procedure until ATTEN TEST cable calibration is complete.

Note

A tool is available for use in converting between the following: dBm, VPP, Watts, and Divisions (Divisions is used with 1.5T systems only). This tool is located on the Signa Release 8.X Service Methods CD-ROM under the Tools directory, titled: SCOPECAL.xls.

Path:

[START] [Programs] [Windows Explorer] [Mrservice (drive)] [Tools] [SCOPECAL.xls].

There is no associated oscilloscope bandwidth limitation (frequency roll-off) for 1.0T systems using a 100 MHz oscilloscope, therefore, if using the SCOPECAL.xls tool the:

- Set the oscilloscope "Volts/div" control knob to 0.5 VDC setting (so that 8 divisions equals 1VP-P).
- "Volts Peak to Peak value of 1.5T Scope Calibrator waveform" would equal 1.
- "# of minor divisions the waveform covers on oscilloscope would equal 40.

3. At operator work space, select the scan desktop ICON in the desktop control panel, if you have not already done so.
4. At the operator work space, prepare the system for a Head Power Monitor scan using the "Service Protocols" procedure below.

TABLE 4-1
SCAN PRESCRIPTION - HEAD PM CHECKS

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

- A. **[New Pt]**
Id: **geservice** <ENTER>
Name: **pmc**
Weight (Lb.): **300**
Set Patient Protocols to **Service**.
- B. At front enclosure:
Landmark in the Head area—remove any coils.
press **LANDMARK**.
press **MOVE TO SCAN**.
- C. At Patient Protocols – select **other**.
- D. In the protocol field, type **o.23.1**<ENTER> (o=Other, 23.1 =series) to load the head protocol
OR select **[o.23] [Series 1] [Accept]**.
[OK] (if required).
- E. **[Save Series]**.
- F. **[Prepare to Scan]**.
- G. **[Research Operations]**.
[Setup Params]. Set TG to 50. **[Done]**.
- H. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **See appropriate Table** <ENTER>.
[Accept].
- I. **[Research Operations] [Download]**.
- J. Select **[Manual Prescan]**.

5. Perform head checks per Table 4-2 (1.5T) or Table 4-3 (1.0T).

Note

Selecting Setpoint Values - All setpoint values are selected by modifying control variables. Instructions are provided in the Scan Conditions column in Table 4-2 (1.5 T).

Note

Wattmeter Reading - Wattmeter reading must be multiplied by cable attenuation factor of cables between the ERBTEC RF amplifier and the wattmeter.

Note

Oscilloscope Reading - Oscilloscope reading must be multiplied by dummy load cable(s) attenuation factor between the ERBTEC RF amplifier and the oscilloscope.

Note: Wattmeter reading must be multiplied by Cable Attenuation factor.

Note 2: If necessary, increase TG until both SENSE and both HEAD LEDs come on. No more than 500 Watts (56.99 dBm) should be necessary.

Note 3: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.

TABLE 4-2
HEAD CHECKS (1.5T) LX

Test	Scan Conditions	Adjustments (Note 1)	Verify	Conclusion
HEAD PEAK POWER <i>high</i>	[modify CVs] calmode = 2 trig = 7 aset = 120 [Accept] [Download] [Manual Prescan]	Increase Transmit Gain (TG) until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure power.	Power is within specifications: Min: 1318 W or 61.2 dBm Max: 1784 W or 62.5 dBm Nom: 1551 W or 61.9 dBm • Pass • Fail	Set TG to 0, Place MONITOR INTERLOCK (MON INTLK) switch to RESET , then MODE1, [Done] .
HEAD PEAK POWER <i>LOW</i>	[modify CVs] aset = 30 [Accept] [Download] [Manual Prescan]	Increase TG until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure Power.	Power is within specifications: Min: 330 W or 55.2 dBm Max: 446 W or 56.5 dBm Nom: 388 W or 55.9 dBm • Pass • Fail	Set TG to 0, Place MON INTLK switch to RESET , then MODE1, [Done] .

HEAD PULSE WIDTH (PW) <i>high</i> (min Limit)	[Modify CVs] calmode = 1 p1 = 4750 aset = 255 pwset = 100 [Accept] [Download] [Manual Prescan]	Increase TG (from 0) until Power measures 200W (53.0 dBm) to 300W (54.8 dBm). (Note 2)	SENSE LEDs (both ON) HEAD LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
HEAD PW <i>High</i> (Max Limit)	[Modify CVs] p1 = 5250 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.
HEADPW <i>LOW</i> (Min Limit)	[Modify CVs] p1 = 400 pwset = 10 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) HEAD LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
HEAD PW <i>LOW</i> (Max Limit)	[Modify CVs] p1 = 600 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

HEAD DUTY CYCLE (DC) HIGH CYCLE (DC) (Min Limit)	[Modify CVs] t3 = 33333 TR_SLOP = 0 calmode = 3 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) HEAD LEDS (both ON) FAULT LEDs (both OFF) (Note 3) • Pass • Fail	[Done].
HEAD DC <i>High</i> (Max limit)	[Modify CVs] p3 = 4767 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 3) • Pass • Fail	[Done]. MON INTLK to RESET , then MODE1.
HEAD DC <i>LOW</i> (Min Limit)	[Modify CVs] p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON), HEAD LEDs (both ON), FAULT LEDs (both OFF) (Note 3) • Pass • Fail	[Done].
HEAD DC <i>LOW</i> (Max Limit)	[Modify CVs] p3 = 917 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 3) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

Note

Selecting Setpoint Values - All setpoint values are selected by modifying control variables. Instructions are provided in the Scan Conditions column in Table 4-3 (1.0 T).

Note

Oscilloscope Reading - Oscilloscope reading must be multiplied by dummy load cable(s) attenuation factor between the ERBTEC RF amplifier and the oscilloscope.

Note 1: If necessary, increase TG until both SENSE and both HEAD LEDs come on. No more than 100 Watts (50.0 dBm) should be necessary.

Note 2: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.

TABLE 4-3
HEAD CHECKS (1.0T) LX

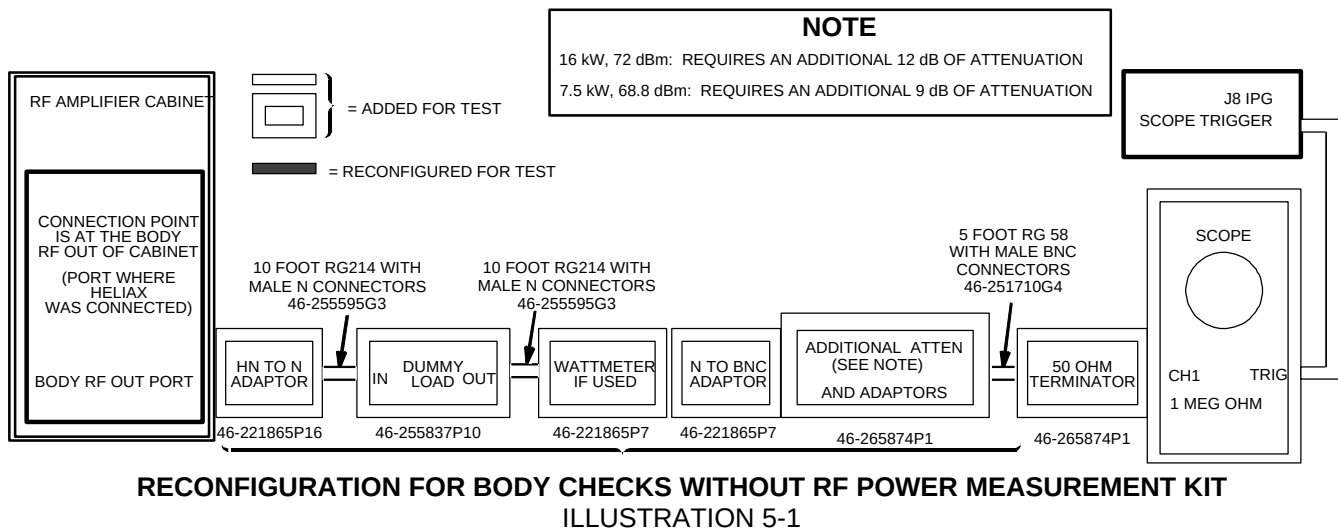
Test	Scan Conditions	Adjustments	Verify	Conclusion
HEAD PEAK POWER <i>high</i>	[Modify CVs] calmode = 2 trig = 7 aset = 50 [Accept] [Download] [Manual Prescan]	Increase Transmit Gain (TG) until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure power.	Power is within specifications: Min: 160 W or 52 dBm Max: 240 W or 53.8 dBm Nom: 200 W or 53 dBm • Pass • Fail	Set TG to 0, Place MONITOR INTERLOCK (MON INTLK) switch to RESET , then MODE1 , [Done] .
HEAD PEAK POWER <i>LOW</i>	[Modify CVs] aset = 30 [Accept] [Download] [Manual Prescan]	Increase TG until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure Power.	Power is within specifications: Min: 100 W or 50 dBm Max: 136 W or 51.34 dBm Nom: 117 W or 50.7 dBm • Pass • Fail	Set TG to 0, Place MON INTLK switch to RESET , then MODE1 , [Done] .

HEAD PULSE WIDTH (PW) <i>high</i> (min Limit)	[Modify CVs] calmode = 1 p1 = 3912 aset = 255 pwset = 35 [Accept] [Download] [Manual Prescan]	Increase TG (from 0) until Power measures 50W (46.9 dBm) to 70W (48.45 dBm). (Note 1)	SENSE LEDs (both ON) HEAD LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
HEAD PW <i>High</i> (Max Limit)	[Modify CVs] p1 = 4324 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.
HEAD PW <i>LOW</i> (Min Limit)	[Modify CVs] p1 = 1076 pwset = 10 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) HEAD LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
HEAD PW <i>LOW</i> (Max Limit)	[Modify CVs] p1 = 1276 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

HEAD DUTY CYCLE (DC) HIGH CYCLE (DC) (Min Limit)	[Modify CVs] t3 = 33333 TR_SLOP = 0 calmode = 3 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) HEAD LEDS (both ON) FAULT LEDs (both OFF) (Note 3) • Pass • Fail	[Done].
HEAD DC <i>High</i> (Max limit)	[Modify CVs] p3 = 4767 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 2) • Pass • Fail	[Done]. MON INTLK to RESET , then MODE1.
HEAD DC <i>LOW</i> (Min Limit)	[Modify CVs] p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) HEAD LEDS (both ON) FAULT LEDs (both OFF) (Note 2) • Pass • Fail	[Done].
HEAD DC <i>LOW</i> (Max Limit)	[Modify CVs] p3 = 917 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 2) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

5- BODY CHECKS

The following generic Illustration 5-1 has been provided for sites without the RF Power Measurement Kit.



Note

Maximum RF Signal input for any scope is 30 dBm (10 Volts Peak, 20 VPP).

Note

Both the dummy load and the cable that connects the amplifier output to the dummy load input must be calibrated per *DUMMY LOAD AND CABLES CALIBRATION*. See Signa Release 8.X Service Methods CD-ROM. If the cable has not been calibrated, STOP; do not continue with this procedure until cable calibration is complete.

Verify the selectable oscilloscope input termination is 1 Meg ohm when using the RF Power Measurement Kit 50 ohm terminator. Otherwise, the oscilloscope should be 50 ohm termination for all RF measurements.

1. **For 1.5T systems:** Connect per Illustration 5-1 to Body output (J13 at I/F panel / J2 at Body Directional Coupler) of the RF/PEN cabinet. If using a wattmeter use the proper element. Or use the RF Power Measurement Kit card # 72 (16 kW, 72 dBm). For 1.5T is necessary to use the 1.5T Scope Calibrator card to compensate for scope bandwidth limitations (frequency roll-off), refer to the Appendix for instructions.

Verify the selectable oscilloscope input termination is 1 Meg ohm when using the RF Power Measurement Kit 50 ohm terminator. Otherwise, the oscilloscope should be 50 ohm termination for all RF measurements.

Note

Calibrate The Amplifier Output To Wattmeter Input Cable - The cable that is used to connect the amplifier output to the wattmeter input must be calibrated per procedure Dummy Load and Cables Calibration. If the cable has not been calibrated, STOP, do not continue with this procedure until ATTEN TEST cable calibration is complete.

Note

A tool is available for use in converting between the following: dBm, VPP, Watts, and Divisions (Divisions is used with 1.5T systems only). This tool is located on the Signa Release 8.X Service Methods CD-ROM under the Tools directory, titled: SCOPECAL.xls.

Path:

[START] [Programs] [Windows Explorer] [Mrservice (drive)] [Tools] [SCOPECAL.xls].

2. **For 1.0T systems:** Connect per Illustration 5-1 to Body output (J13 at I/F panel / J2 at Body Directional Coupler) of the RF/PEN cabinet. If using a wattmeter use the proper element. Or use the RF Power Measurement Kit card # 68.8 (7.5 kW, 68. dBm).

Note

Calibrate The Amplifier Output To Wattmeter Input Cable - The cable that is used to connect the amplifier output to the wattmeter input must be calibrated per procedure Dummy Load and Cables Calibration. If the cable has not been calibrated, STOP, do not continue with this procedure until ATTEN TEST cable calibration is complete.

Note

A tool is available for use in converting between the following: dBm, VPP, Watts, and Divisions (Divisions is used with 1.5T systems only). This tool is located on the Signa Release 8.X Service Methods CD-ROM under the Tools directory, titled: SCOPECAL.xls.

Path:

[START] [Programs] [Windows Explorer] [Mrservice (drive)] [Tools] [SCOPECAL.xls].

There is no associated oscilloscope bandwidth limitation (frequency roll-off) for 1.0Tsystems using a 100 MHz oscilloscope, therefore, if using the SCOPECAL.xls tool the:

- Set the oscilloscope "Volts/div" control knob to 0.5 VDC setting (so that 8 divisions equals 1VP-P).
- "Volts Peak to Peak value of 1.5T Scope Calibrator waveform" would equal 1.
- "# of minor divisions the waveform covers on oscilloscope would equal 40.

3. At operator work space, select the scan icon in the desktop control panel, if you have not already done so.
4. At the operator work space, prepare the system for a Body Power Monitor scan using the "Service Protocols" procedure below.

TABLE 5-1
SCAN PRESCRIPTION - BODY PM CHECKS

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

- A. **[New Series]**
At Patient Protocols – select **other**.
- B. In the protocol field, type **o.23.2<ENTER>** (o=Other, 23.2 =series) to load the body protocol
OR select **[o.23] [Series 2] [Accept]**.
[OK] (if required).
- C. **[Save Series]**.
- D. **[Prepare to Scan]**.
- E. **[Research Operations]**.
[Setup Params]. Set TG to 50. **[Done]**.
- F. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **dcset <ENTER>, 255 <ENTER>**
CV name: **t3 <ENTER>, 20000 <ENTER>**
CV name: **See appropriate Table <ENTER>**.
[Accept].
- F. **[Research Operations] [Download]**.
- G. Select **[Manual Prescan]**.

5. Perform body checks per Table 5-2 (1.5T) or Table 5-3 (1.0T).

Note

Selecting Setpoint Values - All setpoint values are selected by modifying control variables. Instructions are provided in the Scan Conditions column in Table 5-2 (1.5 T).

Note

Wattmeter Reading - Wattmeter reading must be multiplied by cable attenuation factor of cables between the ERBTEC RF amplifier and the wattmeter.

Note

Oscilloscope Reading - Oscilloscope reading must be multiplied by dummy load cable(s) attenuation factor between the ERBTEC RF amplifier and the oscilloscope.

Note 1: Wattmeter reading must be multiplied by Cable Attenuation factor.

Note 2: If necessary, increase TG until both SENSE and both BODY LEDs come on. No more than 5 kW (66.99 dBm) should be necessary.

Note 3: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.

TABLE 5-2
BODY CHECKS (1.5T) LX

Test	Scan Conditions	Adjustments (Note 1)	Verify	Conclusion
BODY PEAK POWER <i>high</i>	[Modify CVs] calmode = 2 trig = 7 aset = 120 [Accept] [Download] [Manual Prescan]	Increase Transmit Gain (TG) until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure power.	Power is within specifications: Min: 8793.9 W or 69.44 dBm Max: 11897.7 W or 70.75 dBm Nom: 10345.8 W or 70.15 dBm • Pass • Fail	Set TG to 0, Place MONITOR INTERLOCK (MON INTLK) switch to RESET , then MODE1 , [Done] .
BODY PEAK POWER <i>LOW</i>	[Modify CVs] aset = 30 [Accept] [Download] [Manual Prescan]	Increase TG until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure Power.	Power is within specifications: Min: 2198.5 W or 63.42 dBm Max: 2974.5 W or 64.73 dBm Nom: 2586.5 W or 64.13 dBm • Pass • Fail	Set TG to 0, Place MON INTLK switch to RESET , then MODE1 , [Done] .

BODY PULSE WIDTH (PW) <i>high</i> (min Limit)	[Modify CVs] calmode = 1 p1 = 4750 aset = 255 pwset = 100 [Accept] [Download] [Manual Prescan]	Increase TG (from 0) until Power measures 2 kW (63.01 dBm) to 3 kW (64.77 dBm). (Note 2)	SENSE LEDs (both ON) BODY LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
BODY PW <i>High</i> (Max Limit)	[Modify CVs] p1 = 5250 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.
BODY PW <i>LOW</i> (Min Limit)	[Modify CVs] p1 = 400 pwset = 10 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) BODY LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
BODY PW <i>LOW</i> (Max Limit)	[Modify CVs] p1 = 600 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

BODY DUTY CYCLE (DC) HIGH CYCLE (DC) (Min Limit)	[Modify CVs] t3 = 33333 TR_SLOP = 0 calmode = 3 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) BODY LEDS (both ON) FAULT LEDs (both OFF) (Note 3) • Pass • Fail	[Done].
BODY DC <i>High</i> (Max limit)	[Modify CVs] p3 = 4767 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 3) • Pass • Fail	[Done]. MON INTLK to RESET , then MODE1.
BODY DC <i>LOW</i> (Min Limit)	[Modify CVs] p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) BODY LEDS (both ON) FAULT LEDs (both OFF) (Note 3) • Pass • Fail	[Done].
BODY DC <i>LOW</i> (Max Limit)	[Modify CVs] p3 = 917 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 2) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

Note

Selecting Setpoint Values - All setpoint values are selected by modifying control variables. Instructions are provided in the Scan Conditions column in Table 5-3 (1.0 T).

Note

Oscilloscope Reading - Oscilloscope reading must be multiplied by dummy load cable(s) attenuation factor between the ERBTEC RF amplifier and the oscilloscope.

Note 1: If necessary, increase TG until both SENSE and both BODY LEDs come on. No more than 1 kW (60 dBm) should be necessary.

Note 2: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.

TABLE 5-3
BODY CHECKS (1.0T) LX

Test	Scan Conditions	Adjustments (Note 1)	Verify	Conclusion
BODY PEAK POWER <i>high</i>	[Modify CVs] calmode = 2 trig = 7 aset = 110 [Accept] [Download] [Manual Prescan]	Increase Transmit Gain (TG) until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure power.	Power is within specifications: Min: 4100 W or 66.127 dBm Max: 4500 W or 66.53 dBm Nom: 4300 W or 66.33 dBm • Pass • Fail	Set TG to 0, Place MONITOR INTERLOCK (MON INTLK) switch to RESET , then MODE1 , [Done] .
BODY PEAK POWER <i>LOW</i>	[Modify CVs] aset = 30 [Accept] [Download] [Manual Prescan]	Increase TG until first FAULT LED on, measure Power. Continue increasing TG until other FAULT LED on, measure Power.	Power is within specifications: Min: 1100 W or 60.41 dBm Max: 1300 W or 61.14 dBm Nom: 1200 W or 60.79 dBm • Pass • Fail	Set TG to 0, Place MON INTLK switch to RESET , then MODE1 , [Done] .

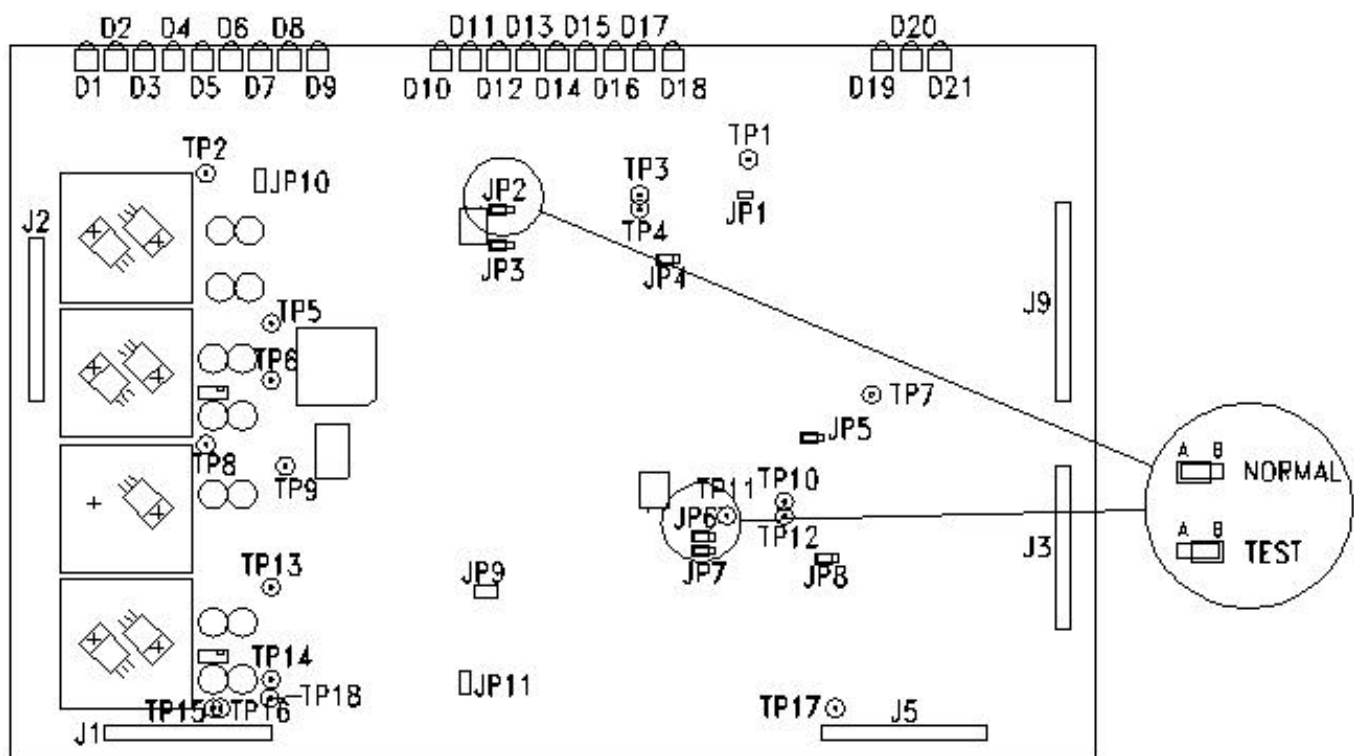
BODY PULSE WIDTH (PW) <i>high</i> (min Limit)	[Modify CVs] calmode = 1 p1 = 3912 aset = 255 pwset = 35 [Accept] [Download] [Manual Prescan]	Increase TG (from 0) until Power measures 500 W (56.99 dBm) to 700 W (58.45 dBm). (Note 2)	SENSE LEDs (both ON) BODY LEDs (both ON) FAULT LEDs (both OFF) • Pass • Fail	[Done].
BODY PW <i>High</i> (Max Limit)	[Modify CVs] p1 = 4324 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done] , MON INTLK to RESET , then MODE1 .
BODY PW <i>LOW</i> (Min Limit)	[Modify CVs] p1 = 1076 pwset = 10 [Accept] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON), BODY LEDs (both ON), FAULT LEDs (both OFF), • Pass • Fail	[Done].
BODY PW <i>LOW</i> (Max Limit)	[Modify CVs] p1 = 1276 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) • Pass • Fail	[Done] , MON INTLK to RESET , then MODE1 .

BODY DUTY CYCLE (DC) HIGH CYCLE (DC) (Min Limit)	[Modify CVs] t3 = 33333 TR_SLOP = 0 calmode = 3 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) BODY LEDS (both ON) FAULT LEDs (both OFF) (Note 2) • Pass • Fail	[Done].
BODY DC <i>High</i> (Max limit)	[Modify CVs] p3 = 4767 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 2) • Pass • Fail	[Done]. MON INTLK to RESET , then MODE1.
BODY DC <i>LOW</i> (Min Limit)	[Modify CVs] p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan]	Do not change TG.	SENSE LEDs (both ON) BODY LEDS (both ON) FAULT LEDs (both OFF) (Note 2) • Pass • Fail	[Done].
BODY DC <i>LOW</i> (Max Limit)	[Modify CVs] p3 = 917 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 2) • Pass • Fail	[Done], MON INTLK to RESET , then MODE1.

6- AMPLIFIER SHUTDOWN VERIFICATION

Tests in this section verify that each power monitor is capable of shutting down the RF Amplifier independently of the other, and further verify both the logic and relay shutdowns for each channel. Setpoints are not checked in this section — this test is forcing a shutdown.

1. It is assumed that the Body output is still connected as in Body Checks section, and that the monitor interlock switch is in MODE 1 position.
2. Place jumpers **JP2 & JP6** on the Communications Manager/Power Monitor Board in the **normal** position, (A). See Illustration 6-1.



COMMUNICATIONS MANAGER/POWER MONITOR BOARD JUMPER SETUP
ILLUSTRATION 6-1

Note

Amplifier Tripped While Moving JP2 - If the amplifier is tripped while moving JP2, IPER (In Place Error Recovery) should reset it about three minutes after the fault condition is cleared. If problems are encountered with the amplifier operation after such a trip, rebooting Signa may be required to restore proper operation.

3. Verify that the Power On and Ready LEDs are illuminated for both power monitors. This confirms that JP2 & JP6 are in the normal position.

4. Disable power monitor B by removing JP9 on the Communications Manager/Power Monitor board.
5. At operator work space, select the scan desktop icon in the desktop control panel, if you have not already done so.
6. At the operator work space, prepare the system for a Amp Shutdown Verification scan using the "Service Protocols" procedure, see below.

TABLE 6-1

SCAN PRESCRIPTION - HIGH VOLTAGE RELAY SHUTDOWN — BODY MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. [New Series]

At Patient Protocols – select **other**.

- B. In the protocol field, type **o.23.3<ENTER>** (o=Other, 23.3 =series) to load the body protocol
OR select **[o.23] [Series 3] [Accept]**.

[OK] (if required).

C. [Save Series]**D. [Prepare to Scan].****E. [Research Operations].**

[Setup Params]. Set TG to 0. **[Done]**.

F. [Research Operations].

[Display CVs]. Highlight CV Name and enter the following:

CV name: **trig <ENTER>, 1 <ENTER>**

CV name: **aset <ENTER>, 120 <ENTER>**

CV name: **calmode <ENTER>, 2 <ENTER>**

CV name: **pwset <ENTER>, 255 <ENTER>**

CV name: **dcset <ENTER>, 255 <ENTER>**

CV name: **p1 <ENTER>, 3100 <ENTER>**

CV name: **p3 <ENTER>, 2400 <ENTER>**

F. [Research Operations] [Download].**G. Select [Manual Prescan].**

7. Increase transmit gain (TG) until wattmeter reads 0.5 to 1 kW. The Sense and Body LEDs should be on for power monitor A only. If other indication is observed, verify that all power monitor related sense cables are properly connected, not swapped.
8. Continue increasing TG until monitor A Fault LED lights. Quickly verify the following:
 - Fault LED on for monitor A (resets automatically after two seconds).
 - RF amplifier is shutdown and displaying fault code 84 or 85 (IPER resets the amplifier 10 to 15 seconds after the power monitor fault clears, i.e., a few seconds after the Fault LED goes off on the amplifier). Software Message Log - Software will log an 84 or 85 fault in the message log.
 - Verify oscilloscope or Wattmeter (press FWD PEP then MAX) reads zero RF power.

Note

IPER Resets The RF Amplifier - Do not perform any operations on the screen or keyboard while IPER is resetting the RF amplifier.

9. While waiting for the RF amplifier to recover, move jumper JP8 from the *A* position to the *B* position.
10. After IPER has reset the RF amplifier, first decrease TG by 50 counts (5 dB), then press START SCAN on operator work space to initiate looping.
11. Increase TG until monitor A Fault LED illuminates. Quickly verify the following:
 - RF amplifier is shutdown and displaying fault code 84 or 85 (IPER resets the amplifier 10 to 15 seconds after the power monitor fault clears, i.e., a few seconds after the Fault LED goes off on the amplifier). Software Message Log - Software will log an 84 or 85 fault in the message log.
 - Verify oscilloscope or Wattmeter (press FWD PEP then MAX) reads zero RF power.

Note

IPER Resets The RF Amplifier - Do not perform any operations on the screen or keyboard while IPER is resetting the RF amplifier.

12. While waiting for the amplifier to recover:
 - a) Move jumper JP8 from the *B* position to the *A* position.
 - b) Reconnect power monitor B by replacing JP9 on the Communications Manager/Power Monitor Board.
 - c) Disconnect power monitor A by removing JP1 on the Communications Manager/Power Monitor Board.
13. After IPER has reset the RF amplifier, first decrease TG by 50 counts (5 dB), then press START SCAN on the operator work station to initiate looping.
14. Increase TG until monitor B Fault LED illuminates. Quickly verify the following:
 - Fault LED on monitor B control board (resets automatically after six seconds).

- RF amplifier is shutdown and displaying fault code 84 or 85 (IPER resets the amplifier 10 to 15 seconds after the power monitor fault clears, i.e., as soon as the Fault LED goes off on the amplifier). Software Message Log - Software will log an 84 or 85 fault in the message log.
- Verify oscilloscope or Wattmeter (press FWD PEP then MAX) reads zero RF power.

Note

IPER Resets The RF Amplifier - Do not perform any operations on the screen or keyboard while IPER is resetting the RF amplifier.

15. After IPER has reset the RF amplifier, select **[Back Up]**.

16. **[Set Up Parameters]** set TG to 0 (zero).

17. **[Cancel]** or **[End]** current study.

- a) Re-connect power monitor A by installing JP1 on the Communications Manager/Power Monitor Board.

7- DECLARATION FORM PREPARATION

1. Prepare *Direction 2147006, Signa Power Monitor Functional Test Declaration*. Refer to Data Sheet for Power Monitor Test Declaration Form.
2. File completed declaration form per current Company Policy.

8- SYSTEM RESTORATION

1. Replace JP9 and JP1 on the Communications Manager/Power Monitor Board. See Illustration 3-4.
2. Return the rocker switch on the front panel of the RFSC Module to OFF, no red will be visible.
3. Disconnect dummy load, wattmeter, and test equipment from rear of RF cabinet.
4. Reconnect head and body heliax cables to rear of RF cabinet.
5. Ensure that JP2 and JP6 are in the Normal position, (A), on the Communications Manager/Power Monitor Board. See Illustration 3-4.
6. Ensure that Power On and Ready LEDs are illuminated for both power monitors.
7. Verify that jumper JP8 is in position A.
8. Close the top lid of the RF system controller power supply module. Properly screw RFSC Module cover in place.
9. Remove the variable attenuator from the J103 / J105 RFSC Module RF path. Reconnect BNC cables to RFIN, MR1A15J105 and EFB RFOUT, MR1A15J103. See Illustration 3-1.
10. Perform one satisfactory head scan.
11. Ensure that Sense and Head LEDs for both power monitors are on during head scan.
12. Perform one satisfactory body scan.
13. Ensure that Sense and Head LEDs for both power monitors are on during body scan.
14. Properly screw RFSC Module to RF/Pen rails.
15. Remove key from the monitor interlock switch, and replace the front door on the RF/PEN cabinet.
16. Perform power monitor functional check Data Sheet Direction 2147006, "Declaration" form prepared and filed according to current Company Policy.

APPENDIX A —SIGNAL POWER MONITOR FUNCTIONAL TEST DECLARATION FORM

Signa Release 8.X RF/PEN 1.0T/1.5T CABINET — 2147006

SITE																																	
(NAME) _____	(SYSTEM ID, ROOM NUMBER, ETC., IF APPLICABLE) _____																																
(STREET) _____	(SYSTEM CABINET SERIAL NUMBER) _____																																
(CITY, STATE, ZIP CODE) _____	(CURRENT SYSTEM SOFTWARE RELEASE) _____																																
TEST EQUIPMENT																																	
(BIRD WATTMETER SERIAL NO.) _____	(OSCILLOSCOPE SERIAL NO.) _____	(RF POWER MEASUREMENT KIT SERIAL NO.) _____																															
(DUMMY LOAD PLUS CABLES ATTEN FACTOR) _____		(AMP TO WATTMETER CABLE ATTEN FACTOR) _____																															
PERFORMANCE																																	
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left; padding: 2px;">FUNCTION CHECK STATUS</th> <th style="text-align: center; padding: 2px;">PASS</th> </tr> </thead> <tbody> <tr><td style="padding: 2px;">BODY PEAK POWER TEST</td><td style="text-align: center; padding: 2px;">☐</td></tr> <tr><td style="padding: 2px;">BODY PULSE WIDTH TEST</td><td style="text-align: center; padding: 2px;">☐</td></tr> <tr><td style="padding: 2px;">BODY DUTY CYCLE TEST</td><td style="text-align: center; padding: 2px;">☐</td></tr> <tr><td style="padding: 2px;">HEAD PEAK POWER TEST</td><td style="text-align: center; padding: 2px;">☐</td></tr> <tr><td style="padding: 2px;">HEAD PULSE WIDTH TEST</td><td style="text-align: center; padding: 2px;">☐</td></tr> <tr><td style="padding: 2px;">HEAD DUTY CYCLE TEST</td><td style="text-align: center; padding: 2px;">☐</td></tr> <tr><td style="padding: 2px;">SHUTDOWN VERIFICATION</td><td style="text-align: center; padding: 2px;">☐</td></tr> </tbody> </table>	FUNCTION CHECK STATUS	PASS	BODY PEAK POWER TEST	☐	BODY PULSE WIDTH TEST	☐	BODY DUTY CYCLE TEST	☐	HEAD PEAK POWER TEST	☐	HEAD PULSE WIDTH TEST	☐	HEAD DUTY CYCLE TEST	☐	SHUTDOWN VERIFICATION	☐	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th colspan="2" style="text-align: center; padding: 2px;">CONFIGURATION FILE ENTRIES</th> </tr> </thead> <tbody> <tr> <td style="padding: 2px;">Cable loss coefficient</td> <td style="padding: 2px;">Coil loss coefficient</td> </tr> <tr> <td style="padding: 2px;">Head Coil _____</td> <td style="padding: 2px;">_____</td> </tr> <tr> <td style="padding: 2px;">Body Coil _____</td> <td style="padding: 2px;">_____</td> </tr> <tr> <td style="padding: 2px;">AMP Cal Head _____</td> <td style="padding: 2px;">ABSOLUTE AVG SAR _____</td> </tr> <tr> <td style="padding: 2px;">AMP Cal Body _____</td> <td style="padding: 2px;">ABSOLUTE PEAK SAR _____</td> </tr> <tr> <td style="padding: 2px;">Average SAR Factor _____</td> <td style="padding: 2px;">Peak SAR Factor _____</td> </tr> </tbody> </table>			CONFIGURATION FILE ENTRIES		Cable loss coefficient	Coil loss coefficient	Head Coil _____	_____	Body Coil _____	_____	AMP Cal Head _____	ABSOLUTE AVG SAR _____	AMP Cal Body _____	ABSOLUTE PEAK SAR _____	Average SAR Factor _____	Peak SAR Factor _____
FUNCTION CHECK STATUS	PASS																																
BODY PEAK POWER TEST	☐																																
BODY PULSE WIDTH TEST	☐																																
BODY DUTY CYCLE TEST	☐																																
HEAD PEAK POWER TEST	☐																																
HEAD PULSE WIDTH TEST	☐																																
HEAD DUTY CYCLE TEST	☐																																
SHUTDOWN VERIFICATION	☐																																
CONFIGURATION FILE ENTRIES																																	
Cable loss coefficient	Coil loss coefficient																																
Head Coil _____	_____																																
Body Coil _____	_____																																
AMP Cal Head _____	ABSOLUTE AVG SAR _____																																
AMP Cal Body _____	ABSOLUTE PEAK SAR _____																																
Average SAR Factor _____	Peak SAR Factor _____																																
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center; padding: 5px;">RESTORATION CHECK LIST</th> <th style="width: 5%;"></th> </tr> </thead> <tbody> <tr> <td style="padding: 5px;">Replace JP9 and JP1 on the Communications Manager/Power Monitor Board.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">Return rocker switch on front panel of RF System Controller Module to "OFF" position.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">JP2 & JP6 in up position (NORMAL) on Communications Manager/Power Monitor Board.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">POWER ON and READY LEDs ON on both Power Monitors.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">Close top lid of RF System Controller Power Supply Module.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">SENSE and HEAD LEDs ON on for both Power Monitors during head scan.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">Verify jumper JP8 is across pins 2-3.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">SENSE and BODY LEDs ON on for both Power Monitors during body scan.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> <tr> <td style="padding: 5px;">Key removed from MONITOR INTERLOCK switch and cover replaced on RF/Pen Cabinet.</td> <td style="text-align: center; padding: 5px;">☐</td> </tr> </tbody> </table>				RESTORATION CHECK LIST		Replace JP9 and JP1 on the Communications Manager/Power Monitor Board.	☐	Return rocker switch on front panel of RF System Controller Module to "OFF" position.	☐	JP2 & JP6 in up position (NORMAL) on Communications Manager/Power Monitor Board.	☐	POWER ON and READY LEDs ON on both Power Monitors.	☐	Close top lid of RF System Controller Power Supply Module.	☐	SENSE and HEAD LEDs ON on for both Power Monitors during head scan.	☐	Verify jumper JP8 is across pins 2-3.	☐	SENSE and BODY LEDs ON on for both Power Monitors during body scan.	☐	Key removed from MONITOR INTERLOCK switch and cover replaced on RF/Pen Cabinet.	☐										
RESTORATION CHECK LIST																																	
Replace JP9 and JP1 on the Communications Manager/Power Monitor Board.	☐																																
Return rocker switch on front panel of RF System Controller Module to "OFF" position.	☐																																
JP2 & JP6 in up position (NORMAL) on Communications Manager/Power Monitor Board.	☐																																
POWER ON and READY LEDs ON on both Power Monitors.	☐																																
Close top lid of RF System Controller Power Supply Module.	☐																																
SENSE and HEAD LEDs ON on for both Power Monitors during head scan.	☐																																
Verify jumper JP8 is across pins 2-3.	☐																																
SENSE and BODY LEDs ON on for both Power Monitors during body scan.	☐																																
Key removed from MONITOR INTERLOCK switch and cover replaced on RF/Pen Cabinet.	☐																																
_____ (PERFORMED BY - PLEASE PRINT) _____ (SIGNATURE) _____ (DATE)																																	

APPENDIX B — RF POWER MEASUREMENT KIT USE

B-1 RF Formulas and 1.5T Division Conversions

1. Understanding Cal Card process (using 1.5T Scope Calibrator, 4 dBm input):

$$\underline{1.00 \text{ VPP (4.0 dBm)}} = \underline{0.125 \text{ VPP each Major Division}} = 0.025 \text{ VPP each Minor Division}$$

Set to 8 Major Div 5 Minor Divisions

The process instructs the user to set 1VP-P equal to 8 Major oscilloscope divisions. The 8 Major Divisions are equal to 40 Minor Divisions. With this concept understood it can be said that 25.28 Minor Divisions is equal to 5 Major Divisions plus almost one third of 1 Minor Division (or ~ .3 of 1 Minor Division).

2. dBm converted to VP-P to Minor Divisions example:

- dBm [÷] 20 [=] [INV LOG] [X] 0.632 [=] VP-P.

$$0 \text{ dBm} = \frac{0.632 \text{ VPP (= 0dBm)}}{0.025 \text{ VPP each Minor Division}} = \mathbf{25.28 \text{ Minor Divisions}}$$

3. RF Formulas:

VP-P to dBm Calculation:

$$\text{VP-P [÷] 0.632 [=] [LOG] [X] 20 [=] dBm.}$$

dBm to VP-P Calculation:

$$\text{dBm [÷] 20 [=] [INV LOG] [X] 0.632 [=] VP-P.}$$

Example: Using 3.65 dBm, Scope Power reading listed on Kit card

$$3.65 \text{ [÷] 20 [=] (0.1825) [INV LOG] (1.5222991) [X] 0.632 [=] (0.9620931) VP-P.}$$

dBm to Watts Calculation:

$$\text{dBm [÷] 10 [=] [INV LOG] [=] [X] 0.001 [=] Total Watts.}$$

B-2 RF Power Measurement Kit Use with 1.0T Systems — Recalculation

The RF Power Measurement Kit contains a 1.5 Tesla Scope Calibrator. This is a 63.86 MHz oscillator that provides a 4 dBm (1.00 VP-P) output into “final” 50 ohms as measured on an oscilloscope. This 1.5T Scope Calibrator is not to be used with 1.0T systems.

Note

1.0T Sites with RF Power Measurement Kit cards that list “Div” (versus VP-P, mVP-P) for the Scope Reading must perform the re-calculation. Additional RF Formulas have been provided.

1.0T Sites with RF Power Measurement Kit cards that have “Div” (scope reading) listed can bypass this Section but must perform the following dBm (scope power) to VP-P calculation:

$\text{dBm} [\div] 20 [=] [\text{INV LOG}] [X] 0.632 [=] \text{VP-P}.$

Example: Using 3.65 dBm, “Scope Power” reading listed on Kit card

$3.65 [\div] 20 [=] (0.1825) [\text{INV LOG}] (1.5222991) [X] 0.632 [=] (0.9620931) \text{VP-P}.$

APPENDIX C — POWER MONITOR MEASUREMENTS

This Appendix Section will guide the user through a series of power monitor measurements and calculations. The tutorial examples are specific to the Head Checks Power Monitor Tables for 1.5T or 1.0T, however, the “process” described can be modified for use with body as well. The tutorial examples listed will assume the use of an oscilloscope and the RF Power Measurement Kit, however, if a Kit is not available the Attenuation value used during hardware setup will be substituted. For first time users it is important to look over the entire Appendix C before using it to avoid confusion.

Section C-1 guides the user through a series of fill-in-the-blank steps. Table C-1 is provided as an additional tutorial example. Sub-Section C-1-1 is not required, but has been provided to allow the user to determine the VP-P acceptable range.

Section C-2 is more involved. Because a range of dBm is required the user will need to perform several calculations for the overall “Minimum”, “Maximum”, and “Note” acceptable range values.

1.5T Sites ONLY — using oscilloscopes less than 300 MHz requires the use of the 1.5T Scope Calibrator tool which is part of the RF Power Measurement Kit. These sites will refer to Appendix B and Appendix D and can convert the measured Minor Divisions to a VP-P (Volts Peak-Peak) value. 1.5T sites using oscilloscopes less than 300 MHz and not performing the oscilloscope display characterization process using the 1.5T Scope Calibrator may get erroneous oscilloscope measurement values. The SCOPECAL.xls program is also available for converting between dBm and Watts or Watts and dBm.

It is helpful to understand that 1 dB is equal to 10 TG ($1/10 \text{ dB} = 1 \text{ TG}$).

C-1 Tutorial example for 1.5T and 1.0T Head Test (Peak Power High):

1. Record the Attenuation Value. This is the Attenuation Value used during the hardware setup using the Dummy Load and Cable Calibration procedure. If using RF Power Measurement Kit, the “Atten” Value is listed in dB on the 1.5T Card #63 or the 1.0T Card #58 on the sticker affixed to the card.

Attenuation Value: _____dB

2. Perform first Head Test (“Scan Conditions” column) for Peak Power High:

Measure the Volts Peak to Peak value on the oscilloscope. Record the value below.

Actual VP-P Value: _____VP-P

3. Perform the following calculations:

- A. To convert the Volts Peak to Peak value to units of dBm:

VP-P [÷] 0.632 [=] [LOG] [X] 20 [=] dBm

Actual VP-P Value converted to dBm: _____dBm

- B. To obtain the final dBm value that was measured:

dBm (step 3A) [+] dB (step 1) [=] Final dBm

Final dBm Value: _____dBm

4. Table C-1 is provided as an additional example. Table C-1 uses an arbitrary Attenuation value. The “Verify” column values are directly from the 1.5T and 1.0T Head Peak Power Checks (High), Tables 4-2 or 4-3. Table C-1 describes one set of possible outcomes for the 1.5T and 1.0T Head Peak Power Checks (High). The “process” will be used for the 1.5T and 1.0T Head and Body Peak Power Checks (Low) and Body Peak Power Checks (High), however, each sites final values will be different.

TABLE C-1

EXAMPLE FOR POWER MONITOR FUNCTIONAL HEAD CHECKS, PEAK POWER — HIGH

Process	1.5T Peak Power - High, Table 4-2	1.0T Peak Power - High, Table 4-3
Attenuation in dB	-59.45 dB	-55.47 dB
“Verify” column — Min Range	61.2 dBm	52 dBm
“Verify” column — Max Range	62.5 dBm	53.8 dBm
Min dBm - Attenuation in dB = Min dBm range	61.2 dBm - 59.45 dB = 1.75 dBm	52 dBm - 55.47 dB = -3.47 dBm
Max dBm - Attenuation in dB = Max dBm range	62.5 dBm - 59.45 dB = 3.05 dBm	53.8 dBm - 55.47 dB = -1.67 dBm
Convert Min dBm range value to VP-P, (step 3A formula): Acceptable Min VPP Range Value	0.773 VP-P	0.424 VP-P
Convert Max dBm range value to VP-P, (step 3A formula): Acceptable Max VPP Range Value	0.898 VP-P	0.521 VP-P

C-1-1 Optional: VPP Ranges for 1.5T and 1.0T Head Test (Peak Power *High* or *Low*):

1. Perform the following calculations to determine VP-P acceptable range from the Min and Max dBm values listed in the “Verify” column per Table 4-2 or 4-3. Refer to Table C-1 for clarification of the process:

Minimum dBm Value $[÷] 20 [=] [\text{INV LOG}] [X] 0.632 [=]$ Min VP-P

Min VP-P Range Value: _____ VP-P

Maximum dBm Value $[÷] 20 [=] [\text{INV LOG}] [X] 0.632 [=]$ Max VP-P

Max VP-P Range Value: _____ VP-P

C-2 Tutorial example for 1.5T and 1.0T Head Test (Pulse Width PW High, Min Limit):

1. Review the Example per Table C-2. The “Adjustments” column values are directly from the 1.5T and 1.0T Head Pulse Width Checks (High, Min Limit), Tables 4-2 or 4-3. Table C-2 describes one set of possible outcomes for the 1.5T and 1.0T Head Pulse Width Checks (High, Min Limit), however, each sites final values will be different.
2. Record the Attenuation Value. This is the Attenuation Value used during the hardware setup using the Dummy Load and Cable Calibration procedure. If using RF Power Measurement Kit, the “Atten” Value is listed in dB on the 1.5T Card #63 or the 1.0T Card #58 on the sticker affixed to the card.

Attenuation Value: _____dB

3. Perform the following calculations to determine dBm acceptable range:

- A. To obtain the Minimum value in dBm:

dBm (Adjustments Column, Table 4-2 or 4-3) [-] **dB** (step 2) [=] **Min dBm**

Minimum dBm Value: _____dBm

- B. To obtain the Maximum value in dBm:

dBm (Adjustments Column, Table 4-2 or 4-3) [-] **dB** (step 2) [=] **Max dBm**

Maximum dBm Value: _____dBm

- C. To obtain the Note value (upper range) in dBm:

dBm (Note Table 4-2 or 4-3) [-] **dB** (step 2) [=] **Note dBm**

Note dBm Value: _____dBm

4. Perform the following calculations to determine VP-P acceptable range:

Minimum dBm Value [÷] 20 [=] **[INV LOG] [X] 0.632** [=] **Min VP-P**

Min VP-P Range Value: _____VP-P

Maximum dBm Value [÷] 20 [=] **[INV LOG] [X] 0.632** [=] **Max VP-P**

Max VP-P Range Value: _____VP-P

Note dBm Value [÷] 20 [=] **[INV LOG] [X] 0.632** [=] **Note Max VP-P.**

Note VP-P Range Value: _____VP-P

5. Review the Example per Table C-2. The Table describes one set of possible outcomes for the 1.5T and 1.0T Head Pulse Width Checks (High, Min Limit).

TABLE C-2

EXAMPLE FOR POWER MONITOR FUNCTIONAL HEAD CHECKS, P.W. HIGH — MIN LIMIT

Process	1.5T P.W. High - Min Limit, Table 4-2	1.0T P.W. High - Min Limit, Table 4-3
Attenuation in dB	-59.45 dB	-55.47 dB
Adjustments Column — Min dBm	53 dBm, 200 W	46.9 dBm, 50 W
Adjustments Column — Max dBm	54.8 dBm, 300 W	48.45 dBm, 70 W
Adjustments Column — Note dBm	56.99 dBm, 500 W	50 dBm, 100 W
Adjustments Column Min dBm - Attenuation in dB = Min dBm (step 3A to convert)	53 dBm - 59.45 dB = -6.45 dBm	46.9 dBm - 55.47 dB = -8.57 dBm
Adjustments Column Max dBm - Attenuation in dB = Max dBm (step 3B to convert)	54.8 dBm - 59.45 dB = -4.65 dBm	48.5 dBm - 55.47 dB = -6.97 dBm
Adjustments Column Note dBm - Attenuation in dB = Note dBm (step 3C to convert)	56.99 dBm - 59.45 dB = -2.46 dBm	50 dBm - 55.47 dB = -5.47 dBm
Min dBm = Min VPP Range Value (step 4A to convert)	-6.45 dBm = 0.301 VP-P	-8.57 dBm = 0.236 VP-P
Max dBm = Max VPP Range Value (step 4B to convert)	-4.65 dBm = 0.370 VP-P	-6.97 dBm = 0.283 VP-P
Note dBm = Note VPP Range Value (step 4C to convert)	-2.46 dBm = 0.476 VP-P	-5.47 dBm = 0.337 VP-P
Acceptable Min to Max VPP Range Acceptable Note VPP Upper Range	0.301 to 0.370 VP-P Note: 0.476 VP-P	0.236 to 0.283 VP-P Note: 0.337 VP-P

APPENDIX D — CALIBRATE SCOPE USING +4dBm SCOPE CALIBRATOR — 1.5T ONLY

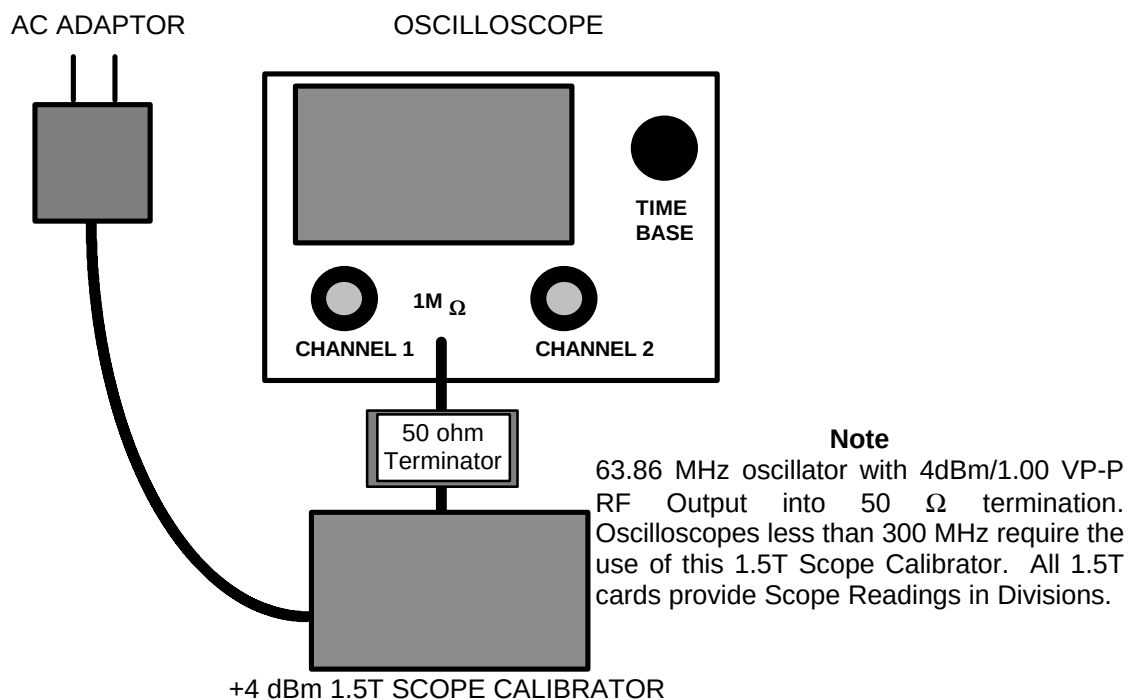
1.5T sites using a 100 MHz oscilloscope (or less than a 300 MHz oscilloscope) must use the RF Power Measurement Scope Calibrator tool to characterize the oscilloscope display. This will compensate for any amplitude error associated with oscilloscope bandwidth limitations or frequency roll-off. The 1.5T Scope Calibrator provides a 63.86 MHz sine wave at +4 dBm/1.00 VPP (as measured on a 300 MHz oscilloscope with 50 ohm termination). To recalculate the 1.5T card divisions reading to VP-P see Appendix B.

1. Verify the Bandwidth Limit button is not selected on the oscilloscope.
2. Verify Channel 1 is set to 1 M ohm input termination.

Note

When measuring RF signals: Use the 50 ohm terminator supplied with the RF Power Measurement Kit and set Channel 1 oscilloscope termination selection to 1 Meg ohm.

3. Connect the 1.5T Scope Calibrator via the 50 ohm terminator to the oscilloscope input. See Illustration D-1.



1.5T SCOPE CALIBRATOR CONNECTIONS
ILLUSTRATION D-1

4. Adjust the channel 1 vertical Volts/div control until amplitude of the waveform is slightly greater than 8 divisions.
5. Adjust the channel 1 vertical Volts/div variable control to achieve exactly 8 divisions.
6. Remove 1.5T Scope Calibrator. Do not remove the 50 ohm terminator attached to the oscilloscope Channel 1 input.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 15, 1998	J.Saperstein	Initial conversion from Toolbook to Word.
1	Nov 12, 1998	M. Keber	Removed obsolete 8.1 information, added data sheet, misc. style guide cleanup.
2	Dec 16, 1998	K. L-P	Fixed Power Monitor Tables. Released on 2187583-1 Rev7 CD-ROM.
3	Jun 23, 1999	R. Lambert	Added note at end of Hardware Prep section notifying location of spreadsheet tool that performs calculations between the following: dBm, VPP, Watts, and Divisions. Added clarification on path to scopecal.xls tool. Added attenuation to EFB bypass. Clarified protocols. Added general Illustrations to Body and Head sections. Added Appendix B and C. Corrected Tables per Bay Tech. Also, corrected all dBm value for Body Peak Power Low for 1.0T ONLY, first 2 test incorrectly showed the same dBm values. Removed Illustrations 4-2, 2-4, 5-2, 5-3. Added additional notes for SCOPECAL.xls Tool. Added measurement example/tutorial. Added info to Amp Shutdown Verification. Released on 2160623-1 Rev8 CD-ROM.